

$$\theta = \frac{a + a^\dagger}{\sqrt{2\omega}} \quad \begin{aligned} a|n\rangle &= \sqrt{n}|n-1\rangle \\ a^\dagger|n\rangle &= \sqrt{n+1}|n+1\rangle \end{aligned}$$

$$\langle i|\theta|j\rangle = \frac{1}{\sqrt{2\omega}} \langle i|a + a^\dagger|j\rangle$$

$$\theta_{ij} = \frac{1}{\sqrt{2\omega}} (\langle i|\sqrt{j}|j-1\rangle + \langle i|\sqrt{j+1}|j+1\rangle)$$

$$= \frac{1}{\sqrt{2\omega}} (\delta_{i,j-1}\sqrt{j} + \delta_{i,j+1}\sqrt{j+1})$$

$$\theta_{ij}^2 = \left(\frac{1}{\sqrt{2\omega}}\right)^2 (\delta_{i,j-1}\sqrt{j} + \delta_{i,j+1}\sqrt{j+1})^2$$

$$\delta_{i,j+1}\delta_{i,j-1} = 0 \quad \delta_{i,j}^2 = \delta_{i,j}$$

$$= \frac{1}{2\omega} (\delta_{i,j-1}j + \delta_{i,j+1}(j+1))$$

$$\theta_{ij}^4 = \left(\frac{1}{\sqrt{2\omega}}\right)^4 (\delta_{i,j-1}\sqrt{j} + \delta_{i,j+1}\sqrt{j+1})^4$$

$$\delta_{i,j}^4 = \delta_{i,j}^3 = \delta_{i,j}^2 = \delta_{i,j} \quad \delta_{i,j+1}\delta_{i,j-1} = 0$$

$$= \frac{1}{4\omega^2} (\delta_{i,j-1}j^2 + \delta_{i,j+1}(j+1)^2)$$

$$\langle i | \hat{O} | j \rangle = \frac{1}{\sqrt{2\omega}} \begin{matrix} 0 & 1 & 2 & 3 \\ j \longrightarrow \\ \begin{matrix} 0: \\ 1: \\ 2: \\ 3: \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 0 & \dots \\ 1 & 0 & \sqrt{2} & 0 & \dots \\ 0 & \sqrt{2} & 0 & \sqrt{3} & \dots \\ 0 & 0 & \sqrt{3} & 0 & \sqrt{4} & 0 & \dots \\ 0 & 0 & 0 & \sqrt{4} & 0 & \dots \end{pmatrix} \end{matrix}$$

$$\hat{O}^2 = \frac{1}{2\omega} \begin{pmatrix} 0 & 1 & 0 & 0 & \dots & 0 & 0 & \dots \\ 1 & 0 & \sqrt{2} & 0 & 0 & 0 & 0 & \dots \\ 0 & \sqrt{2} & 0 & \sqrt{3} & 0 & 0 & 0 & \dots \\ 0 & 0 & \sqrt{3} & 0 & \sqrt{4} & 0 & 0 & \dots \\ 0 & 0 & 0 & \sqrt{4} & 0 & \sqrt{5} & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \quad 2$$

$$= \frac{1}{2\omega} \begin{pmatrix} 1 & 0 & \sqrt{2} & 0 & 0 & 0 & 0 & 0 & \dots \\ 0 & 3 & 0 & \sqrt{6} & 0 & 0 & 0 & 0 & \dots \\ \sqrt{2} & 0 & 5 & 0 & \sqrt{12} & 0 & 0 & 0 & \dots \\ 0 & \sqrt{6} & 0 & 7 & 0 & \sqrt{20} & 0 & 0 & \dots \\ 0 & 0 & \sqrt{12} & 0 & 9 & 0 & \sqrt{30} & 0 & \dots \\ 0 & 0 & 0 & \dots & \dots & \dots & \dots & \dots & \dots \end{pmatrix}$$

~~$\frac{\sqrt{j(j+1)}}{\sqrt{j(j+1)}}$~~ $\frac{\sqrt{j(j+1)}}{\sqrt{j(j+1)}}$ $2i+1$ $\sqrt{j(j+1)}$

$$\hat{O}^4 = \frac{1}{4\omega^2} \begin{pmatrix} 3 & 0 & \sqrt{72} & 0 & \sqrt{24} & 0 & \dots \\ 0 & 15 & 0 & \sqrt{600} & 0 & \sqrt{120} & \dots \\ \sqrt{72} & 0 & 39 & 0 & 28\sqrt{3} & 0 & \dots \\ 0 & \sqrt{600} & 0 & 75 & 0 & 24\sqrt{5} & \dots \\ \sqrt{24} & 0 & 28\sqrt{3} & 0 & 93 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$